

Hyperbolas

ANSWER KEY



Standard form

- 1 For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:
 - a the vertices $(\pm 4, 0)$.
 - b the foci $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.
 - c the equations of the asymptotes $y = \pm \frac{3}{4}x$.
 - 2 Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.
 $a = 2, c = 3, b^2 = 9 - 4 = 5: \frac{y^2}{4} - \frac{x^2}{5} = 1$.
 - 3 Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.
 - a Write the equation in standard form by completing the square. $4(x + 1)^2 - (y - 2)^2 = 4$, i.e. $(x + 1)^2 - \frac{(y - 2)^2}{4} = 1$.
 - b Identify the conic and state its center. A hyperbola opening left-right with center $(-1, 2)$.
-

Identifying conics

- 4 Identify each conic section from its equation:
 - a $x^2 + y^2 = 49$ Circle.
 - b $\frac{x^2}{9} + \frac{y^2}{25} = 1$ Ellipse.
 - c $y^2 - 4x^2 = 16$ Hyperbola.
 - d $y = 3x^2 - 6x + 1$ Parabola.
 - 5 A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.
 $a = 3$ and $\frac{b}{a} = 2$ gives $b = 6: \frac{x^2}{9} - \frac{y^2}{36} = 1$.
-

Hyperbolas centered away from the origin

- 6 For the hyperbola $\frac{(y + 1)^2}{4} - \frac{(x - 3)^2}{9} = 1$, find:
 - a the center $(3, -1)$
 - b the vertices $(3, -1 \pm 2) = (3, 1), (3, -3)$
 - c the direction of opening opens up/down
 - 7 Complete the square to write $9x^2 - 4y^2 - 18x - 16y - 43 = 0$ in standard form, and state the center.
 $9(x^2 - 2x) - 4(y^2 + 4y) = 43: 9(x - 1)^2 - 9 - 4(y + 2)^2 + 16 = 43$, so $9(x - 1)^2 - 4(y + 2)^2 = 36$,
i.e. $\frac{(x - 1)^2}{4} - \frac{(y + 2)^2}{9} = 1$. Center: $(1, -2)$.
-

Eccentricity and applications

- 8 Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

$c = \sqrt{9 + 16} = 5$; $e = \frac{5}{3} \approx 1.67$. Since $e > 1$ and relatively large, the branches are wide and open quickly, resembling straight lines more than a tight curve.

- 9 A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

$2a = 120$, so $a = 60$; $2c = 200$, so $c = 100$; $b^2 = c^2 - a^2 = 10000 - 3600 = 6400$. Equation:
 $\frac{x^2}{3600} - \frac{y^2}{6400} = 1$.

Graphing a hyperbola branch

- 10 The upper-right branch of $y^2 - x^2 = 4$ (as $y = \sqrt{4 + x^2}$) is shown. Use the graph to estimate y at $x = 3$, then verify algebraically, and compare to the asymptote $y = x$.

From the graph, $y \approx 3.6$ at $x = 3$. Algebraically: $y = \sqrt{4 + 9} = \sqrt{13} \approx 3.61$, which is close to but above the asymptote value $y = x = 3$, consistent with the branch approaching but never touching its asymptote.

The hyperbola's defining property

- 11 State the geometric (difference-of-distances) definition of a hyperbola, and use it to verify the vertex $(4, 0)$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) satisfies it.

A hyperbola is the set of points where the difference of distances to the two foci is constant ($= 2a$). At $(4, 0)$: distance to $(5, 0)$ is 1; distance to $(-5, 0)$ is 9. Difference $= 9 - 1 = 8 = 2a = 2(4) \checkmark$.

Comparing all four conic sections

- 12 State the key algebraic feature (coefficient signs) that distinguishes a circle, ellipse, parabola, and hyperbola in a general second-degree equation $Ax^2 + By^2 + \dots = 0$.

Circle: $A = B$ (same sign, equal). Ellipse: A, B same sign, unequal. Parabola: exactly one of A, B is zero (only one variable squared). Hyperbola: A, B opposite signs.

Finding the equation from asymptotes and a point

- 13 A hyperbola centered at the origin has asymptotes $y = \pm \frac{2}{3}x$ and passes through $(3, 0)$. Find its equation. Since it passes through $(3, 0)$, it opens left-right with $a = 3$. From $\frac{b}{a} = \frac{2}{3}$: $b = 2$. Equation: $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

Applications: LORAN navigation

- 14** Two LORAN radio towers are 260 km apart. A ship determines the difference in distances to the two towers is 100 km. Write the equation of the hyperbola of possible ship locations (centered at the midpoint, opening along the line connecting the towers).

$$2a = 100 \Rightarrow a = 50; 2c = 260 \Rightarrow c = 130; b^2 = c^2 - a^2 = 16900 - 2500 = 14400. \text{ Equation: } \frac{x^2}{2500} - \frac{y^2}{14400} = 1.$$